

Deep Learning Ensemble with ESD-Net in Breast Cancer Classification on Mammogram

Ho Dat Tran
Can Tho University, Can Tho
Vinh Long University of Technology Education, Vinh Long
datth@vlute.edu.vn

Thuong Cang Phan
Can Tho University
Can Tho, Vietnam
ptcang@cit.ctu.edu.vn

Vinh Phong Nguyen
Can Tho Oncology Hospital
Can Tho, Vietnam
dr.vinhphong86@gmail.com

Anh Cang Phan*
Vinh Long University of Technology Education
Vinh Long, Vietnam
cangpa@vlute.edu.vn

Abstract—Early diagnosis and treatment of cancer play a critical role in improving patient recovery outcomes. In recent years, deep learning models have demonstrated significant effectiveness in the medical domain, supporting physicians in enhancing diagnostic and treatment capabilities. However, predictions from individual models often lack stability and are highly dependent on input data characteristics, which may compromise classification performance. This paper proposes ESD-Net, an ensemble framework that integrates seven convolutional neural networks (InceptionV3, ResNet152, ResNet50V2, DenseNet121, DenseNet201, VGG19, and Xception) for robust breast cancer classification on mammography images. Prior to training, raw images undergo preprocessing steps including denoising, normalization, and quality enhancement. Predictions from the individual models are then combined using ensemble techniques based on voting strategies to leverage their complementary strengths. The proposed method improves classification performance, achieving 95.8% accuracy with Soft Voting and 95.6% with Hard Voting on the test dataset. These findings highlight the reliability of the proposed model in early breast cancer detection.

Index Terms—mammography, breast cancer, deep learning ensemble.

I. INTRODUCTION

Breast cancer is one of the most prevalent diseases with a high incidence and mortality rate worldwide. According to GLOBOCAN [1], in 2022 there were an estimated 19.9 million new cancer cases globally and nearly 10 million deaths, of which breast cancer in women accounted for 2.2 million cases (23.8%). Mammography plays a crucial role in effective treatment by enabling timely detection of abnormalities. It is widely used as a routine screening technique, particularly for women with suspected clinical symptoms.

In recent years, deep learning models have shown remarkable potential in medical image classification. Ensemble approaches that combine multiple models have emerged as a promising strategy to enhance classification accuracy and reduce bias. For example, Eroğlu et al. [2] proposed a hybrid CNN-based framework using ultrasound images for breast cancer diagnosis, integrating features extracted from AlexNet, MobileNetV2, and ResNet50. The best features were selected using the mRMR method, and classification was performed with Support Vector Machine (SVM) and k-Nearest Neighbor

(k-NN), yielding an accuracy of 95.6%. Similarly, Adedigba [3] evaluated multiple deep learning models (InceptionV3, ResNet50, AlexNet) and an ensemble CNN-TV-SVM approach on a small dataset, achieving 93.6% accuracy. Despite promising results, the small dataset size posed risks of overfitting when generalized to larger datasets. In another study, Hashim [4] developed an ensemble method using Hard Voting for mammography classification by combining Decision Tree, Logistic Regression, and SVM. To address data imbalance, Random Oversampling (ROS) and Standard Scaler normalization were applied. The proposed model achieved 98.25% accuracy with Hard Voting and an average accuracy of 97.38% across 10-fold cross-validation, demonstrating the effectiveness of carefully designed Hard Voting ensembles. Hanis et al. [5] conducted a meta-analysis evaluating machine learning models for breast cancer classification using digital mammography and tomosynthesis. By analyzing 36 studies with 68 models and data from public databases (MIAS, DDSM) as well as primary datasets, they reported an overall high diagnostic accuracy with an AUC of 0.90, sensitivity of 0.83, and specificity of 0.84. Factors influencing performance included country, data source, and model type, with neural network-based models and U.S. datasets performing best. However, limitations included heterogeneity among studies and insufficient tomosynthesis data. These findings confirm the potential of machine learning in breast cancer screening.

Building upon these studies, we propose ESD-Net, a robust ensemble model designed to exploit the advantages of diverse architectures to improve reliability in mammography-based cancer classification. Careful parameter tuning and model selection ensure strong performance, while evaluation metrics—including accuracy, loss, F1-score, precision, and recall—validate the effectiveness of the proposed approach.

II. RELATED WORK

A. Breast cancer

Breast cancer develops when cells in breast tissue grow abnormally and uncontrollably. In diagnostic practice, breast lesions are typically classified into two main categories: benign and malignant. Benign lesions are usually non-life-threatening, grow slowly, are non-invasive, and exhibit clearly defined boundaries. In contrast, malignant lesions are charac-

*Corresponding author

terized by rapid growth, potential invasion into surrounding tissues, and the ability to metastasize to other organs, posing a serious threat to life if not detected and treated promptly. Accurate distinction between these two lesion types is a critical factor in selecting appropriate treatment strategies and improving clinical outcomes.

Large-scale clinical studies have demonstrated that regular mammographic screening programs can reduce breast cancer mortality rates by 20–40%, while significantly improving early-stage detection. Nevertheless, diagnostic accuracy remains influenced by image quality and breast tissue characteristics, particularly in women with dense breast tissue. In this context, the demand for artificial intelligence (AI)-based computer-aided diagnostic (CAD) systems is increasing, aiming to minimize errors, improve consistency, and enhance screening efficiency.

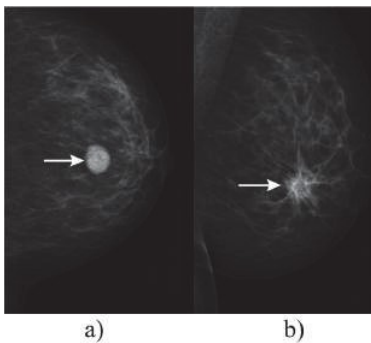


Fig. 1: Example mammography images of (a) benign and (b) malignant breast tumors.

B. Deep learning models

InceptionV3 [6] is an advanced version of the Inception architecture, derived from GoogLeNet but significantly optimized in terms of feature extraction and computational efficiency. A notable improvement is the introduction of factorized convolutions, replacing large filters (e.g., 5×5 or 7×7) with sequences of smaller filters ($1 \times n$ and $n \times 1$). This approach reduces the number of parameters while preserving the ability to learn complex features. Additionally, InceptionV3 employs Batch Normalization to accelerate convergence and mitigate overfitting, alongside optimized pooling operations for spatial reduction. Its architecture is well-suited for multi-class classification tasks, balancing depth and performance. ResNet152 [7] and ResNet50V2 [8] are representative variants of the Residual Network (ResNet) architecture, addressing the vanishing gradient problem in very deep networks through skip connections. ResNet152, with 152 layers, uses bottleneck residual blocks (1×1 , 3×3 , 1×1 convolutions) to reduce dimensionality before applying 3×3 convolutions, thereby saving memory. This design is suitable for large-scale datasets such as ImageNet due to its ability to capture hierarchical features. ResNet50V2 improves upon the original ResNet by restructuring the placement of Batch Normalization and skip connections. Instead of applying BN after each convolution, ResNet50V2 employs a pre-activation scheme (BN before activation), enabling

more stable gradient flow and higher accuracy, while maintaining efficiency for real-time tasks. DenseNet121 [9] and DenseNet201 [10] are based on dense connectivity, where each layer receives input from all preceding layers via concatenation. This design encourages feature reuse, reduces parameters through shared representations, and mitigates the vanishing gradient problem by providing multiple gradient propagation paths. DenseNet121 (121 layers) is well-suited for resource-constrained tasks requiring high accuracy, while DenseNet201 (201 layers) provides more detailed feature learning, particularly valuable in medical imaging tasks such as lesion or X-ray classification. VGG19 [11] is a classical deep architecture with 19 layers, known for its simplicity, consisting of repeated 3×3 convolutional layers and 2×2 max-pooling layers. Its main advantage lies in structural uniformity, where the exclusive use of 3×3 filters enables learning of local-to-global features (edges, textures, and complex objects). While VGG19 provides stable performance with approximately 144 million parameters, it requires substantial computational resources. It is often used as a pre-trained backbone for tasks such as image segmentation or style transfer. Xception [12] (Extreme Inception) extends the Inception framework by replacing conventional convolutions entirely with depthwise separable convolutions. This mechanism applies filters independently to each channel, reducing parameters significantly compared to traditional convolutions while maintaining strong classification performance.

C. Deep learning ensembles

A Deep Learning Ensemble [13] combines multiple neural network architectures to improve classification accuracy and stability. By leveraging the complementary strengths of different architectures—such as the diverse feature extraction of InceptionV3, the stability of ResNet, and the parameter efficiency of DenseNet—ensembles reduce variance, limit overfitting, and enhance generalization capability. Two common strategies for combining outputs are Hard Voting and Soft Voting.

Hard Voting [14] is a simple and widely used ensemble method, particularly in classification problems. It operates based on majority voting: each base model independently predicts a label (benign or malignant), and the final decision is determined by the majority vote among all models. Soft

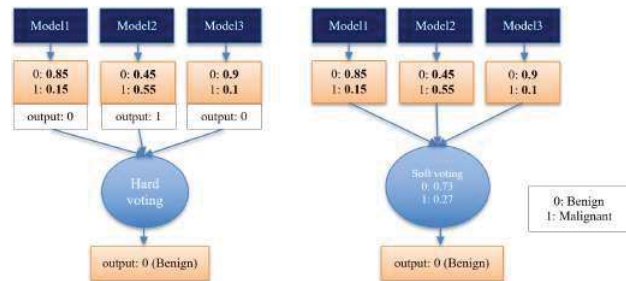


Fig. 2: Illustration of Hard Voting vs. Soft Voting mechanisms.

Voting [14] is a more advanced ensemble strategy, applied when base models provide class probabilities or confidence scores rather than discrete labels. The method averages the

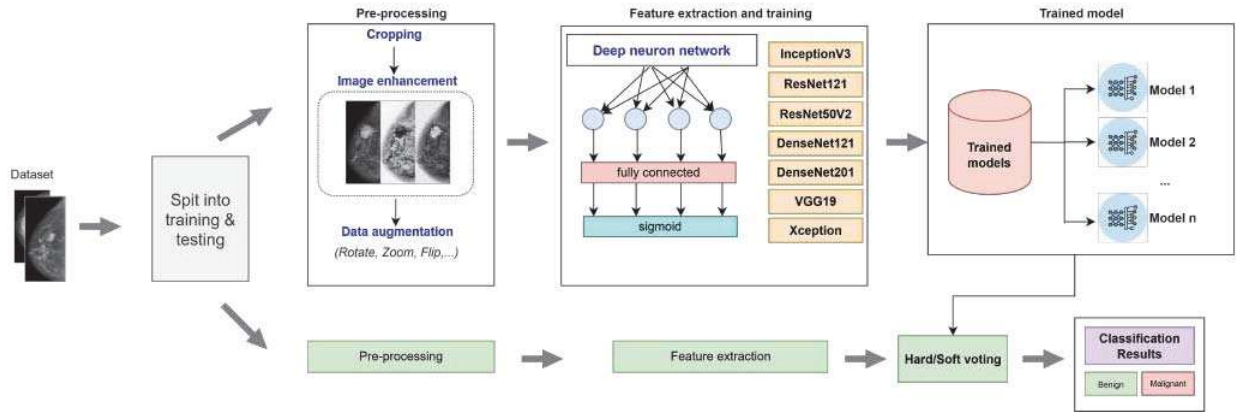


Fig. 3: Proposed ESD-Net model for cancer classification on mammograms.

predicted probabilities across models to determine the final class label. The general formula for Soft Voting is expressed as:

$$P_{\text{soft}}(z = c | l) = \frac{1}{k} \sum_{i=1}^k P_i(z = c | l) \quad (1)$$

where: k : number of classifiers, $P_i(z = c | l)$: probability predicted by the i^{th} classifier that sample l belongs to class c .

D. Evaluation Metrics

We employ several commonly used machine learning metrics to evaluate the performance of breast cancer detection and classification on mammograms. These include accuracy, sensitivity, model loss during training, precision, recall, and F1-score. The equations used are as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

$$F1 - \text{score} = 2 \times \frac{\text{Pre.} \times \text{Rec.}}{\text{Pre.} + \text{Rec.}} \quad (5)$$

$$\text{Loss}(C, P) = - \sum_i C_i \log(P_i) \quad (6)$$

$$mAP = \frac{1}{n} \sum_{i=1}^n AP_i \quad (7)$$

Where, TP: The quantity of benign samples accurately recognized by the model; TN: Quantity of malignant specimens accurately recognized by the model; FP: Quantity of benign samples erroneously classified as malignant by the model; FN: Count of malignant samples erroneously classified as benign by the model. C_i : Accurate designation of the i^{th} class; P_i : The probability of class i as predicted by the model; n : the quantity of classes; AP_i : Average precision for class i .

III. PROPOSED METHOD

The proposed ESD-Net framework for mammography-based breast cancer classification, illustrated in Fig.3, consists of two main stages: (1) Model Training – incorporating image preprocessing and data augmentation techniques to enhance learning capability, while optimizing neural network architectures tailored for mammography characteristics. (2) Testing and Evaluation – deploying the trained models on experimental datasets to assess accuracy, sensitivity, and specificity in distinguishing between benign and malignant breast tumors.

Training Stage: During training, input images are pre-processed using techniques such as cropping, image enhancement, and augmentation (e.g., flipping, rotation, zooming). The processed images are then fed into the proposed machine learning models for feature extraction and training. The dataset is trained independently across seven deep learning models: InceptionV3, ResNet152, ResNet50V2, DenseNet121, DenseNet201, VGG19, and Xception. Each model is trained separately on the same dataset with unified hyperparameters. Early stopping is employed to mitigate overfitting. The best-performing weights of each model are saved for subsequent testing.

Testing Stage: Test images undergo the same preprocessing and feature extraction pipeline, after which Hard Voting and Soft Voting ensemble techniques are applied to generate the final classification output (benign or malignant).

IV. EXPERIMENTAL RESULTS

A. Implementation environment and dataset

Implementation environment: The proposed model was implemented on the Kaggle platform, using TensorFlow and Keras libraries for model training. The experiments were conducted on a system equipped with an Intel Xeon® 2.00 GHz processor, 30 GB RAM, and an NVIDIA Tesla P100 GPU with 16 GB VRAM.

Dataset: Images were collected from the publicly available VinDr dataset [15] and from several hospitals in Can Tho City. All data were validated by medical specialists before inclusion in training and testing. The dataset was divided into

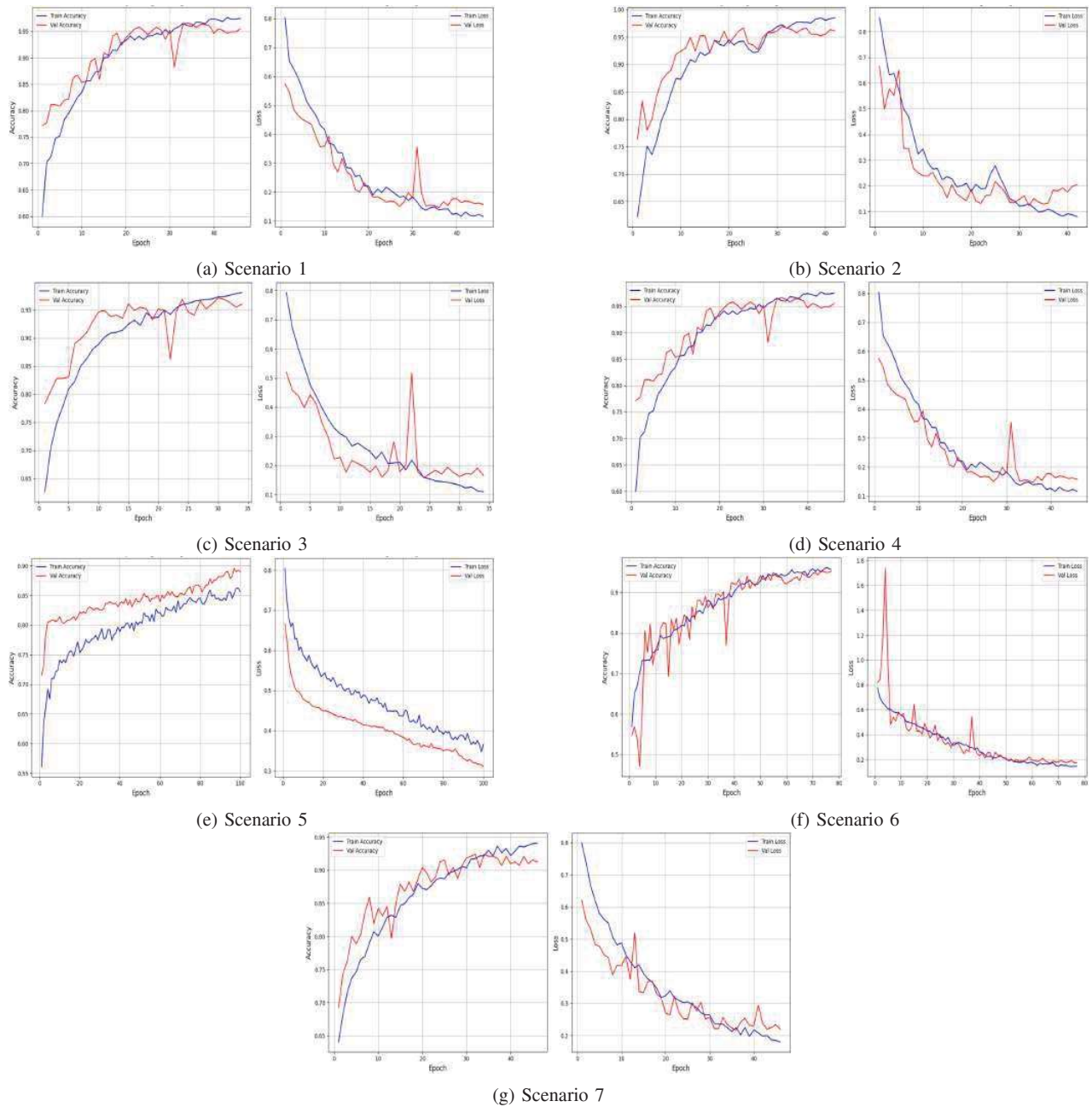


Fig. 4: Results of accuracy and loss measurements of scenarios.

three subsets: training (2,841 images), validation (355 images), and testing (1,068 images), encompassing two classes: benign and malignant.

B. Training results

Fig.4 illustrates the accuracy and loss curves across epochs for the seven experimental scenarios. Overall, accuracy increased while loss decreased consistently with training, indicating effective model learning. Most scenarios achieved high accuracy and low loss, reflecting strong generalization. However, Sce. 3 exhibited noticeable fluctuations and divergence between training and validation curves, resulting in lower overall performance.

Fig.5 present classification performance on the test dataset. Among the scenarios, Sce. 1 achieved the highest accuracy (96.5%), followed by Sce. 4 (95.6%) and Sce. 5 (95.1%). These models demonstrated strong capability in learning discriminative features for benign versus malignant lesions. In terms of loss, Sce. 4 achieved the lowest loss (0.184), followed by Sce. 5 (0.190) and Sce. 1 (0.196). Conversely, Sce. 3 recorded the highest loss (0.387), consistent with its unstable training performance.

C. Training time comparison

Fig.6 summarizes training times across models, showing significant differences. DenseNet201 required the longest

TABLE I: Model training scenarios and parameters

Scenarios	Network Architecture	Dimension (H×W)	Batch Size	Epochs	Optimizer	Learning Rate
1	InceptionV3	384×192	32	100	Adam	1e-4
2	Resnet152	384×192	32	100	Adam	1e-4
3	Resnet50V2	384×192	32	100	Adam	1e-4
4	Densenet121	384×192	32	100	Adam	1e-4
5	Densenet201	384×192	32	100	Adam	1e-4
6	VGG19	384×192	32	100	Adam	1e-4
7	Xception	384×192	32	100	Adam	1e-4
8	ESD-Net – Hard voting					
9	ESD-Net – Soft voting					

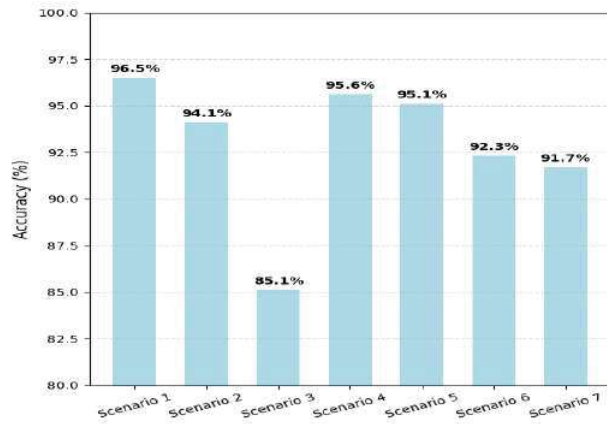


Fig. 5: Chart of accuracy assessment scenarios.



Fig. 6: Chart of training time scenarios.

training time (156.52 minutes), followed by VGG19 (136.75 minutes) and ResNet152 (81.63 minutes). This reflects the correlation between model depth, architectural complexity, and computational cost. In contrast, ResNet50V2 (52.27 minutes) and DenseNet121 (67.38 minutes) trained considerably faster, owing to their lightweight and optimized designs. These findings suggest that for real-world applications with limited resources, models with shorter training times may be preferable. Overall, training time analysis plays a critical role in balancing classification performance against computational cost when selecting models for practical deployment.

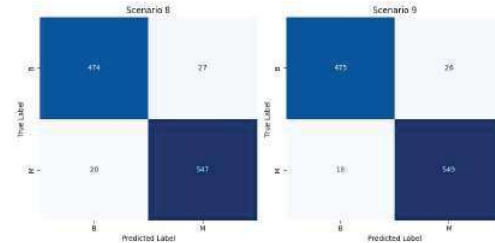


Fig. 7: Confusion matrix on two scenarios 8, 9.

D. Confusion matrix on the test dataset

Through the analysis of the confusion matrix of each scenario, it can be seen that most of the models achieve high correct classification rates in both classes. Sce. 2 shows that the ResNet152 model stands out when correctly classifying up to 471 Benign samples and 550 Malignant samples, with a very low confusion rate of only 30 Benign samples and 17 Malignant samples. Sce. 1, 3 and 4 also show good performance, with the number of confused samples ranging from 20 to 40, reflecting the stable feature learning ability. In contrast, Sce. 5, the DenseNet201 model clearly shows its weakness with the highest number of confusions (102 Benign samples and 57 Malignant samples), indicating that this architecture has difficulty in clearly distinguishing between the two groups on the research dataset. Xception and VGG19 of Sce. 6 and 7 maintain average classification performance; Although Xception has a high number of mistakes in the Benign class (59 samples), it still correctly classifies the majority of Malignant samples, while VGG19 has a more balanced number of mistakes between the two classes.

E. Ensemble model results

Based on the outputs of the individual deep learning models, ensemble techniques were applied using Hard Voting and Soft Voting. The performance of these ensembles was evaluated in terms of accuracy and discriminative capability between benign and malignant tumors, with results summarized through their corresponding confusion matrices. The results of Fig.7 show that both ensemble methods give good prediction results with high accuracy, in which Sce. 9 shows higher performance than Sce. 8. Specifically, Sce. 8 correctly predicts 475 benign samples and 549 malignant samples, while Sce. 9 predicts 474 and 547 respectively. At the same time, the number of falsely malignant samples is reduced from 20 to 18, showing that Sce. 9 has higher sensitivity to malignant classification cases - which is especially important in the problem of supporting early cancer diagnosis.

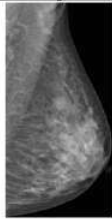
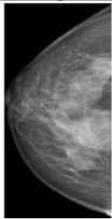
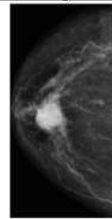
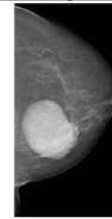
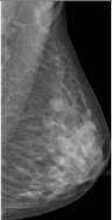
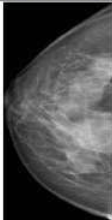
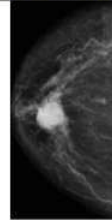
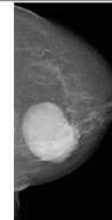
Scenarios	Results			
	Img 1	Img 2	Img 3	Img 4
Hard voting				
	Actual label: Benign Predicted: Benign Voting: 9/7 Malignant 7/7 Benign	Actual label: Benign Predicted: Benign Voting: 6/7 Malignant 7/7 Benign	Actual label: Malignant Predicted: Malignant Voting: 7/7 Malignant 0/7 Benign	Actual label: Malignant Predicted: Malignant Voting: 7/7 Malignant 0/7 Benign
Soft voting				
	Actual label: Benign Predicted: Benign $P_{soft(z=0)}: 1.00$ $P_{soft(z=1)}: 0.00$	Actual label: Benign Predicted: Benign $P_{soft(z=0)}: 0.99$ $P_{soft(z=1)}: 0.01$	Actual label: Malignant Predicted: Malignant $P_{soft(z=0)}: 0.00$ $P_{soft(z=1)}: 1.00$	Actual label: Malignant Predicted: Malignant $P_{soft(z=0)}: 0.01$ $P_{soft(z=1)}: 0.99$

Fig. 8: Results displayed on test dataset.

F. Classification performance evaluation

Based on the experimental results, the deep learning ensemble methods yielded notable improvements in classification metrics such as Precision, Recall, F1-score, and mAP. Across all scenarios (1–9), the models achieved relatively high accuracy, ranging from 85.1% to 95.8%. Among these, Sce. 9 obtained the highest accuracy of 95.8%, with both Precision and Recall exceeding 95% for benign (B) and malignant (M) classes, highlighting the model's stability in classification. The Hard voting method is effective in cases where there is a high level of agreement between models. However, for difficult images where some models predict incorrectly, this method is prone to bias if the incorrect model is the majority. This shows a limitation of the Hard voting method which is that it does not consider the confidence (probability) of each model, but simply relies on the number of votes. Meanwhile, the Soft voting method is more effective because it combines the probabilities from multiple models to help reduce the influence of out-of-phase predictions from each individual model, thereby improving the stability of the classification results.

V. CONCLUSION

In this paper, we proposed the ESD-Net model to improve the efficiency of breast cancer classification on mammograms through the combination of deep learning models. Specifically, the output of the models is combined using two combination strategies: Hard voting and Soft voting. Experimental results show that the proposed model outperforms the single models in terms of classification performance. In particular, Soft voting achieves the highest accuracy of 95.8%, slightly better than Hard voting (95.6%), and gives better F1-score and Recall results in both benign and malignant classes. The high efficiency of Soft voting can be explained

TABLE II: Classification results for class B (Benign) and M (Malignant) on the test dataset.

Sce.	Class	Precision (%)	Recall (%)	F1-score (%)	mAP (%)
1	B	94.6	94.6	92.0	94.7
	M	94.2	94.9	91.0	
2	B	96.5	94.0	95.3	95.6
	M	94.8	97.0	95.9	
3	B	94.3	95.4	94.8	95.1
	M	95.9	94.9	95.4	
4	B	95.3	92.0	93.6	94.1
	M	93.2	95.9	94.5	
5	B	87.5	79.6	83.4	85.1
	M	83.3	90.0	86.5	
6	B	90.2	93.8	92.0	92.3
	M	94.3	91.0	92.6	
7	B	92.5	88.2	90.3	91.1
	M	90.0	93.7	91.8	
8	B	95.9	94.6	95.2	95.6
	M	95.3	96.4	95.8	
9	B	96.3	94.8	95.5	95.8
	M	95.4	96.8	96.1	

by the ability to comprehensively exploit probabilistic information from multiple deep learning models, helping to minimize random errors and increase stability in predictions. In particular, the model shows a clear advantage in detecting malignant cancer cases - a key factor in supporting early diagnosis and timely treatment. In addition, the difference in performance between the ensemble scenarios shows that the classification results also depend on the characteristics of each component model as well as the input data. This opens up the next research direction to optimize the selection of the appropriate background model and ensemble strategy for each specific situation. In the future, we will continue to extend this approach to the problem of tumor segmentation on mammograms, moving towards building a comprehensive diagnostic support system that combines both detection and

classification to improve the effectiveness of breast cancer screening and treatment.

REFERENCES

- [1] B. S. Chhikara and K. Parang, "Global Cancer Statistics 2022: the trends projection analysis," 2023.
- [2] Eroğlu, Yeşim, Muhammed Yildirim, and Ahmet Çinar. "Convolutional Neural Networks based classification of breast ultrasonography images by hybrid method with respect to benign, malignant, and normal using mRMR." *Computers in biology and medicine* 133 (2021): 104407.
- [3] Adedigba, Adeyinka P., Steve A. Adeshina, and Abiodun M. Aibinu. "Performance evaluation of deep learning models on mammogram classification using small dataset." *Bioengineering* 9.4 (2022): 161.
- [4] Hashim, Mohammed S., and Ali A. Yassin. "Breast Cancer Prediction Using Soft Voting Classifier Based on Machine Learning Models." *IAENG International Journal of Computer Science* 50.2 (2023).
- [5] Hanis, Tengku Muhammad, Md Asiful Islam, and Kamarul Imran Musa. "Diagnostic accuracy of machine learning models on mammography in breast cancer classification: a meta-analysis." *Diagnostics* 12.7 (2022): 1643.
- [6] Szegedy, Christian, et al. "Rethinking the inception architecture for computer vision." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2016.
- [7] He, Kaiming, et al. "Deep residual learning for image recognition." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2016.
- [8] Szegedy, Christian, et al. "Inception-v4, inception-resnet and the impact of residual connections on learning." *Proceedings of the AAAI conference on artificial intelligence*. Vol. 31. No. 1. 2017.
- [9] Huang, Gao, et al. "Densely connected convolutional networks." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017.
- [10] G. Huang, Z. Liu, G. Pleiss, L. Van Der Maaten, and K. Q. Weinberger, "Convolutional networks with dense connectivity," *IEEE transactions on pattern analysis and machine intelligence*, vol. 44, no. 12, pp. 8704–8716, 2019.
- [11] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *arXiv preprint arXiv:1409.1556*, 2014.
- [12] Chollet, François. "Xception: Deep learning with depthwise separable convolutions." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017.
- [13] Dietterich, Thomas G. "Ensemble methods in machine learning." *International workshop on multiple classifier systems*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2000.
- [14] Hastie, Trevor, Robert Tibshirani, and Jerome Friedman. "The elements of statistical learning." (2009).
- [15] Pham, Hieu Huy, H. Nguyen Trung, and Ha Quy Nguyen. "Vindr-mammo: A large-scale benchmark dataset for computer-aided detection and diagnosis in full-field digital mammography." *Sci Data* (2022).